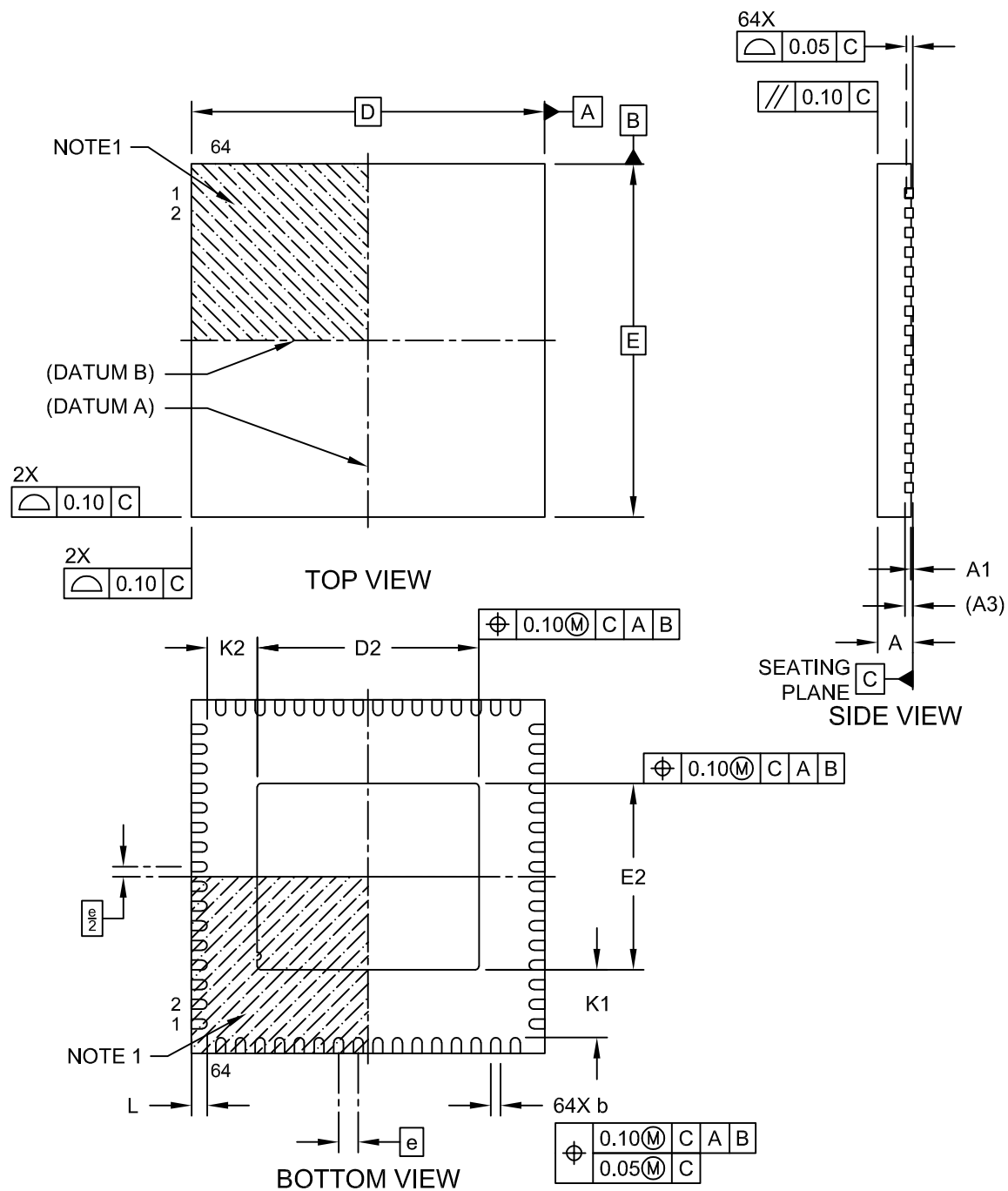


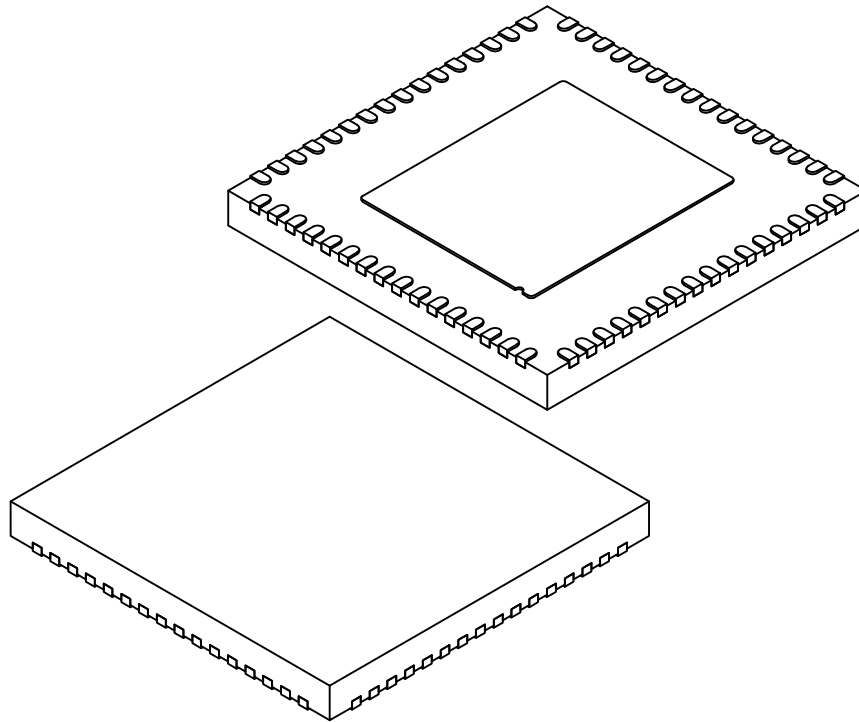
**64-Lead Very Thin Plastic Quad Flat, No Lead Package (TAB) - 9x9x0.9 mm Body [VQFN]  
With 5.65x4.75 mm Exposed Pad; Atmel Legacy GPC ZRA**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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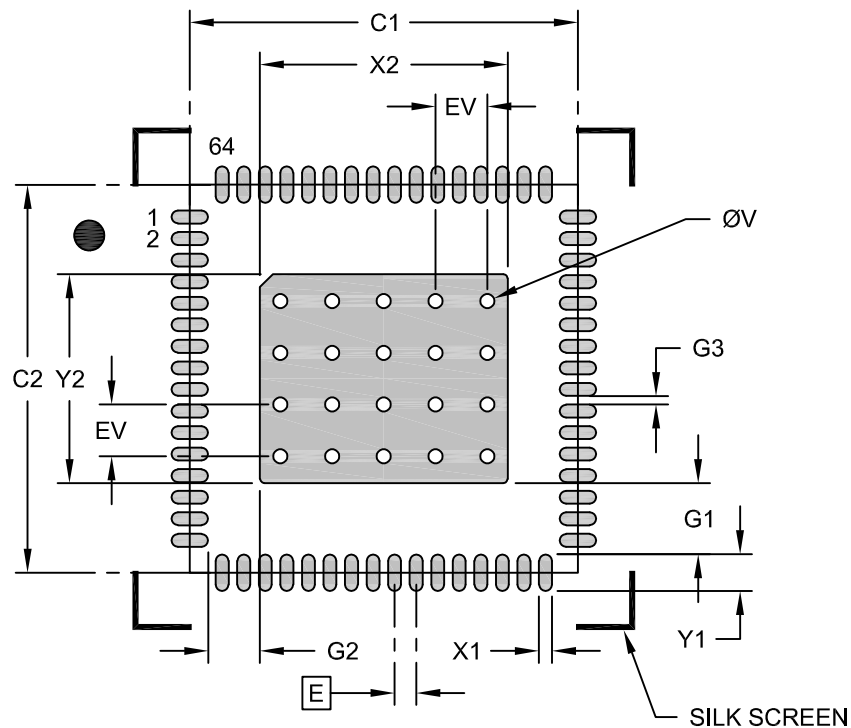
|                         |    | Units | MILLIMETERS |      |      |
|-------------------------|----|-------|-------------|------|------|
| Dimension Limits        |    |       | MIN         | NOM  | MAX  |
| Number of Terminals     | N  |       | 64          |      |      |
| Pitch                   | e  |       | 0.50 BSC    |      |      |
| Overall Height          | A  |       | -           | -    | 0.90 |
| Standoff                | A1 |       | 0.00        | 0.02 | 0.05 |
| Terminal Thickness      | A3 |       | 0.20 REF    |      |      |
| Overall Length          | D  |       | 9.00 BSC    |      |      |
| Exposed Pad Length      | D2 |       | 5.55        | 5.65 | 5.75 |
| Overall Width           | E  |       | 9.00 BSC    |      |      |
| Exposed Pad Width       | E2 |       | 4.65        | 4.75 | 4.85 |
| Terminal Width          | b  |       | 0.18        | 0.25 | 0.30 |
| Terminal Length         | L  |       | 0.35        | 0.40 | 0.45 |
| Terminal-to-Exposed-Pad | K1 |       | 1.72 REF    |      |      |
| Terminal-to-Exposed-Pad | K2 |       | 1.27 REF    |      |      |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

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With 5.65x4.75 mm Exposed Pad; Atmel Legacy GPC ZRA**

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**RECOMMENDED LAND PATTERN**

| Units                            |    | MILLIMETERS |      |      |
|----------------------------------|----|-------------|------|------|
| Dimension Limits                 |    | MIN         | NOM  | MAX  |
| Contact Pitch                    | E  | 0.50 BSC    |      |      |
| Optional Center Pad Width        | X2 |             |      | 5.75 |
| Optional Center Pad Length       | Y2 |             |      | 4.85 |
| Contact Pad Spacing              | C1 |             | 8.90 |      |
| Contact Pad Spacing              | C2 |             | 8.90 |      |
| Contact Pad Width (Xnn)          | X1 |             |      | 0.30 |
| Contact Pad Length (Xnn)         | Y1 |             |      | 0.85 |
| Contact Pad to Center Pad (X32)  | G1 | 1.65        |      |      |
| Contact Pad to Center Pad (X32)  | G2 | 1.20        |      |      |
| Contact Pad to Contact Pad (X60) | G3 | 0.20        |      |      |
| Thermal Via Diameter             | V  |             | 0.33 |      |
| Thermal Via Pitch                | EV |             | 1.20 |      |

**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23432 Rev A